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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the smallest element
c. Bubbling up the largest element to its correct position
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **2**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It complicates the selection of items
c. It makes the data unsorted
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting makes it harder to search for items
c. Sorting helps in finding duplicates quickly
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **4**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of duplicate keys
b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **5**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Bubbling up the largest element to its correct position with each pass
b. Sorting data in a non-sequential manner
c. Using the divide-and-conquer approach
d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **6**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Searching
b. Duplicates
c. All of the above
d. Selection

The correct answer is: All of the above

Question **7**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It allows for faster searching
c. It increases the number of comparisons needed
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list
b. Inserting data into a list
c. Arranging data in a particular format
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Merge Sort
c. Insertion Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It is less efficient than custom sorting algorithms
c. It sorts data out of the box efficiently
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **12**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Quick Sort
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sorted()
- c. arrange()
- d. sort()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Divide and conquer
- c. Backtracking
- d. Greedy method

The correct answer is:
Divide and conquer

Question **15**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120

b. 200,470,80,150,40,90,300,400,70,120

c. 80,150,200,470,40,90,300,400,70,120

d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120